Agent-Based Interaction Protocols and Topologies for Manufacturing Task Allocation

Mohammad Owliya, Mozafar Saadat, Guiovanni G. Jules, Mahbod Goharian, and Rachid Anane

***Abstract*—This paper focuses on interaction protocols and topologies of multiagent systems (MASs) for task allocation, particularly in manufacturing application. Resource agents in manufacturing are members of a network whose possible logical topologies and governing interaction protocol influence the scheduling and control in the MAS. Four models are presented in this paper, each having specific rules and characteristics for scheduling and task allocation. Two models out of the four use a well-known standard interaction method [contract-net protocol (CNP)], while the others are proposed in this paper. The newly proposed models are based on ring topology and algorithms developed in the research. A Java-based MAS was also developed to simulate different scenarios of task allocation and to compare the four models in terms of some scheduling performance indicators, using cases from manufacturing. The results produced meaningful differences between the four models, including their strengths and weaknesses. Two models, namely, modified ring and CNP-based peer-to-peer, gave superior performance compared with the others. Furthermore, the proposed modified ring exhibits significant potential in handling manufacturing task allocation applications.**

***Index Terms*—Agent-based systems, contract-net protocol (CNP), manufacturing scheduling, task allocation, topology.**

I. Introduction

SCHEDULING is a major decision-making process in many engineering systems. In general, it is to determine when each activity of the system starts and ends. Considering availability and constraints of the system's limited resources, scheduling is essentially task allocation to the resources over time. Many methodologies and techniques have been suggested during the past several decades for this purpose in numerous application areas.

Manufacturing task allocation is, in particular, one of the most difficult and popular scheduling problems. An important scheduling issue in manufacturing is related to job shop, which consists of a set of jobs and a number of machines that handle a maximum of one job at a time. Each job is composed of a set of consecutive operations that are to be processed on a given machine within a given time interval, and without interruption. Job shop scheduling is actually a task allocation problem which seeks the solution that minimizes the total lead time required for the completion of all jobs, and/or satisfies other criteria related to cost, machine utilization, etc. This paper falls into this category of manufacturing scheduling.

The requirements of modern manufacturing systems have led the research efforts toward dynamic and decentralized task allocation techniques including agent-based systems derived from distributed artificial intelligence (DAI). Agents are computational systems capable of acting autonomously in a dynamic environment toward a designed purpose. They communicate with one another and with their environment. A community or network of interacting agents is called a multiagent system (MAS). In MAS, various architectures or organizational patterns may exist, each leading to different roles for agents, different rules and relations between them, and, hence, different ways of achieving objectives [1], [2].

Two basic types of the agents' roles in task allocation are operator (task performer) and manager, while the agents can have both roles simultaneously. Rules and protocols for the agents' interaction, on the other hand, regulate relation between the agents for achieving goals [1]. In turn, the interaction protocols are correlated with topology of the agents' network [3]. Together, they may characterize solution models for the task allocation problem. In other words, different basic topologies and corresponding protocols can be considered for task allocation in MAS.

Given a specific protocol and topology, a fundamental issue is *how they compare in terms of manufacturing performance*. Little work has been reported that deals explicitly with this issue. This paper addresses manufacturing performance by introducing new models for agent-based manufacturing job allocation, hence leading to the enhancement of relevant methods in shop floor automation. Four task allocation models have been considered in this paper, each with a basic topology of MAS network and the corresponding rules and protocols. An agent-based simulation system has been developed to support experimental work and to facilitate comparisons of the performance of the models in an industrial case study involving a large manufacturing shop floor.

The following sections of this paper are structured as follows. Section II gives a review of the respective literature. A brief background of distributed and dynamic scheduling in manufacturing is presented, followed by review on agent-based manufacturing task allocation, as well as interaction protocols and topology issues in the network of agents. Section III describes

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contract-net protocol (CNP) for task allocation in MASs and then explains separately the four topology models used in this paper with their related rules for implementation. The first two models introduced are based on CNP, while the last two models use a new protocol. Section IV includes a brief discussion on possible solutions for simulation of the models, followed by description of the agent-based simulation system developed for the experimental part of the research. Evaluation of the system and the models is presented and discussed in Section V.

II. Background Literature

Scheduling is a decision-making process for assignment of a system's limited resources to the tasks that have to be performed such that the system's goals are achieved [4]. The output of the decision-making process is a schedule which specifies when each task starts and finishes, and to which resource it is allocated. In manufacturing job shop, the purpose of scheduling is to allocate operations (tasks) to the time intervals on the machines (resources) [5]–[7]. Published research works in this field, where each customer order is unique (i.e., make-to-order) [4], show evolution and variations of numerous scheduling methodologies ranging from earlier mathematical techniques to various artificial intelligence systems [8]. A clear-cut categorization of the methods is difficult since many of them have been combined with one another to suggest more efficient techniques for different applications. Nevertheless, the demands and requirements of modern manufacturing have led the research endeavors, in general, toward dynamic and distributed approaches.

A. Dynamic and Distributed Approach to Task Allocation

Manufacturing scheduling is actually a task allocation problem which seeks for an optimized solution that satisfies some given criteria. However, an optimized allocation plan might become obsolete by a simple change or disturbance in the real environment [9]. Therefore, dynamic task allocation and the techniques that support it have been highly researched in the past. In a comprehensive survey on dynamic scheduling, the heuristics, metaheuristics, knowledge-based systems, fuzzy logic, neural networks, hybrid techniques, and MASs have been named as such methods [10]. However, most of such techniques can be used both in dynamic real-time and static deterministic scheduling problems.

On the other hand, using a top–down centralized system for task allocation causes rigidity and confines problem-solving ability in the real world [9], [11], although centralization can provide a consistent global view of the state of the system [10]. Distributed approach to control and scheduling is a way to address the inflexibility of the hierarchical systems. Moreover, task allocation in manufacturing can essentially be seen as a distributed problem from both logical and physical points of view [12] and can benefit from distributed methods that improve its reaction to disturbance and allow for parallel computing [6].

Agent-based systems are able to accommodate both requirements mentioned earlier, and therefore have been employed in many manufacturing system research and development projects [1]. The agent-based approach also possesses a prominent position among other dynamic scheduling techniques [10].

B. Agent-Based Systems for Manufacturing Task Allocation

As the manufacturing sector is currently under pressure for fast response to dynamic variations imposed by the market, MASs are widely used to model the complex manufacturing structures and operations [13]. A typical make-to-order environment in the agent-based context has been reflected in [14], where a business needs to make decisions on time (due date of the order) and price offered to the customers. Rajabinasab and Mansour [15] also studied a job shop problem under dynamic events to compare their proposed MAS with some other scheduling methods. However, multiagent frameworks such as that proposed in [16] can model the manufacturing systems with heterarchical interactions to carry out scheduling and react to the changes and disturbances more swiftly.

Agent technology is a form of DAI. It can make a number of advantages for manufacturing task allocation as follows [9]:

1) parallel computing which causes high efficiency and robustness;
2) facilitating the integration of scheduling with process planning;
3) cooperative scheduling;
4) possibility of real-time dynamic rescheduling;
5) easier connection of shop floor scheduling to the enterprise and supply chain levels;
6) possibility of integration with other intelligent techniques for decision making.

In an agent-based system for task allocation, autonomous task and resource agents interact with each other in order to create a schedule which aims at performance maximization by fulfillment of the tasks as soon as possible [16]. Two forms of interaction among agents could be taken into account [17].

1) *Direct interaction*—Agents directly contact and exchange messages with one another or negotiate to reach an agreement.
2) *Indirect interaction*—Agents have no direct communication, but leave information for others or collect it via their environment anonymously.

In these systems, task allocation can be handled in the following ways or variations of them [17].

1) *Static allocation*: There is a specific agent that can decompose a job and allocate them to other agents. This is a central hierarchical type of decision making.
2) *Predetermined dynamic allocation*: Agents dynamically assume the role of manager and contractors, and each contractor can, in turn, perform as a manager for lower levels of task decomposition. However, a top-level manager still exists in the system. CNP, which will be discussed in more detail later, is in this category.
3) *General dynamic allocation*: There is no top-level predefined manager in this case. Tasks are posed to all agents, and each participant tries to offer a solution by taking the "manager" role; the problem then is finding the "best" one.

For development of an agent-based task allocation system, four main issues must be dealt with [5]: 1) representation of the physical world entities by agents with an explicit relationship between the entity and relevant agent; 2) system architecture and topology of the agents' network; 3) interaction protocols which are closely related to the topology; and 4) decision scheme for individual agents, which is not independent of the interaction protocols. This paper focuses on topology and interaction protocol issues of the agent-based systems, which are reviewed in the next subsection.

From the application viewpoint, the paper by Parunak in 1987 was one of the earliest works in manufacturing task allocation that used agent-based concepts and the CNP for assignment of jobs to machines [18]. A marketlike agent model for resource allocation, allowing multistep negotiation between parts and resources, was later developed [19]. By emergence and development of holonic manufacturing systems in the mid-1990s [20], other research efforts proposed various job/resource assignment methods for manufacturing applications (see [21] for instance). MASs acted as an appropriate technology platform for the holonic concept [22]. Those methods combine central rules with distributed strategies to improve responsiveness, instead of using only central optimized and complex scheduling algorithms [23]. In a further research effort, the agent-based holonic control has been integrated with evolutionary algorithms to support decentralized decision making in distributed scheduling [24].

C. Interaction Protocol and Topology Issues in the Agent-Based Systems

As mentioned earlier, interaction protocols regulate relations between the agents for achieving overall goals in a MAS [1]. Market mechanisms have offered strong protocols for task allocation within agent-based systems which are dominant in this field [1], [25]. Variations of the CNP are the most commonly used ones, although other methods such as auction-, pricing-, yellow-page-, and game-theory-based, among others, exist [17], [26], [27]. The two economically inspired mechanisms for task allocation were compared by Dash *et al.* [28]. Application of the market methods is either task or resource driven, which have virtually the same result in terms of assignment of a series of tasks to a number of resources. It is however argued that market mechanisms have also some drawbacks; for instance, it is hard to guarantee avoidance of extreme situations [1].

Interaction protocols are not independent of topology of the agents' network [3]. Together, they determine how the task allocation is performed. In a MAS as a network of agents, similar to many other networks, agent interaction, collaboration, and data and knowledge sharing depend on the system topology [29]. In a series of studies, three topologies named weblike, starlike, and gridlike were presented and compared by Zhu *et al.* [29], [30]. This was done in an application of MASs with consideration of criteria for communication between the agents, dependence to complete tasks, and sharing knowledge/data. A fourth topology was also presented in the same works, named "hierarchical collective agent network," which combines some features of the other three models. This topology is suitable for the specific application of knowledge-intensive multiagent cooperation [29], [30].

The advantages and disadvantages of the aforementioned topologies in terms of autonomy, adaptation, scalability, and efficiency of cooperation have been assessed by Zhu [3]. In this paper, the applicability of each topology to different environments is discussed. It suggests that a proper topology leads to a better behavior of MAS, and reports the results of evaluating many MAS products or development tools to identify which topology is more common in which categories of applications. The research shows that starlike and weblike topologies are prevalent in task allocation.

Ye *et al.* [31] suggested a task allocation model on a peer-to-peer (P2P; weblike) structure as opposed to the centralized structure. Their approach is appropriate for large-scale networks, where a single agent has a limited number of connections with neighboring agents. In such context, they mainly focused on reallocating tasks through mediators, in the case that neighbor resources are not sufficient to carry out the tasks completely. In manufacturing job shop application, however, small networks of limited resources are dealt with.

In another study, the formation of network based on variations of the CNP is discussed, where the order holon agent asks for proposal, and the resource holon agents bid for execution [32]. However, product holon has been added as an intermediary agent in the network. This configuration does not have a generic application in manufacturing, but makes a chain implementation of the CNP in a network of starlike clusters. The topology of the network is not fixed, and changes over time as the result of the change in the state of tasks and resources [32]. Nevertheless, the basic logical topology which depends to the negotiation and allocation protocol remains the same.

CNP has been used in different other applications. For instance, Kodama *et al.* [33] presented a MAS developed for a power distribution grid, which is based on the CNP. In the system, agents interact to exchange information and build a CNP-based network for protection of the grid. They showed through simulation experiments that the CNP-based agent cooperation system can offer an effective strategy for restoration of the power distribution network in case of an accident.

MASs may have a fixed or changing topology. Many research works consider variability of the agents' network topology in different problems [34]–[37]. Agent coordination and readjustment following changes in the underlying network topology of MAS has been the subject of study in [16]. The proposed algorithms for multiagent task/resource negotiation and allocation consider the distribution of agents within the network, and the topology factor at each time. It should be noted, however, that the research is related to geographical distribution and relocation of agents, which causes alteration in the network topology, rather than logical topologies and interaction protocols in MASs.

In another research, the effect of network topology in agent-based manufacturing has been studied [38]. Here, an infrastructure for coordination of agents in a network-based manufacturing system has been presented.

From a broader viewpoint, topological analysis of agent networks has been a subject of investigation by Zhang *et al.*

[39], [40]. Agent network topologies have been classified by them into three general categories: centralized, decentralized, and hybrid, to complete an earlier work by Minar [41]. It is argued that the topology issue is of high importance in agent communication and cooperation. This issue has been studied in a diverse range of problems. For example, the influence of network structure and topology on the system performance in a MAS for information retrieval has been demonstrated in [42]. In another example, the effects of agent interaction topologies in social science problems have been discussed [43]. None of these works, however, specifically address the issue raised in Section I.

III. Agent Interaction Protocols and Topologies in Task Allocation

A manufacturing system, modeled by agents, is a loosely coupled network of communicating and cooperating production entities [1]. In such a network, the connection method between these entities, together with their interaction rules, significantly affects the functionality of the system. As discussed in Section II, research efforts in dynamic distributed scheduling have widely used market mechanisms, particularly standard CNP or its variations for the allocation of tasks to resources. CNP was proposed as a simple and efficient tool [44], which has been later standardized by the Foundation for Intelligent Physical Agents (FIPA) [45]. CNP is a powerful market-based negotiation tool whose significant characteristics are simplicity and efficiency for assigning tasks to the agents that can communicate with one another by sending and receiving messages in the form of FIPA-agent communication language (ACL) [45]. Fig. 1 shows the FIPA standard protocol, while the following steps are a summary of the CNP task allocation process among the contractor agents (known also as participants) and the manager agent (also known as initiator) [44]:

1) task announcement by initiator;
2) task announcement processing by participant;
3) bidding by participant;
4) bid processing by initiator, and awarding the contract;
5) contract processing, reporting result, and termination.

In its original form, CNP is such that a participant awarded with a contract cannot bid for a new task until it has completed its current task. In a modified version, used in this paper, a participant can bid for a new task prior to completing the current task. This is deemed to be more compatible with the real manufacturing environment. Regardless of these variations, it is evident that, in the CNP type of interaction, there is a central initiator, which is surrounded by participants. This provides the basis for a starlike topology in a network of agents. CNP in a star topology is a widely used model for task allocation.

Lau *et al.* [46] proposed an agent-based model for a certain supply chain task allocation issue that has a distributed nature. They used a modified version of the CNP, thus allowing more P2P interactions for information sharing among the resources in the network of agents. Simulation experiments enabled them to compare the performance of the model with centralized scheduling approach and conventional CNP, and the results showed a relative advantage of their model. In a similar way,

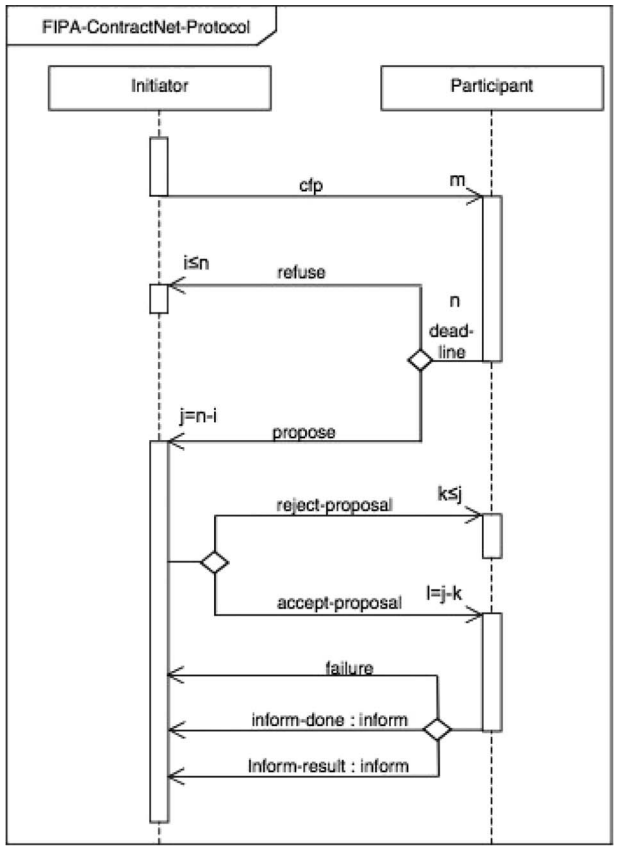


Fig. 1. FIPA contract-net interaction protocol [45].

this research is a comparative study of some feasible models made of topologies and their pertinent interaction protocols in the field of agent-based scheduling.

In this section, the concept of the star model in conjunction with CNP is discussed first. Then, a P2P variation of the CNP-based star model is presented as the second model. The ring topology with the proposed protocol is presented as the third model. The ring model is finally modified and presented as the fourth model to incorporate features of P2P interaction, but with a similar protocol as the ring model. The structure and protocol description of these four models are given in the following subsections. In the development of the four models, the following common manufacturing strategies are used regardless of the topology under consideration:

1) in a series of tasks to be allocated, those in the "critical path" have the highest priority, with the next priority given to the tasks with fewer margins left to due time;
2) task dependence is to be observed—i.e., all prerequisite tasks are carried out prior to the original task;
3) resource agents cannot simultaneously operate on more than one task.

A. Star Model

As already stated, with the CNP in its simplest form (where resource contractors are only connected to the central manager), a star network forms, as shown in Fig. 2. Regardless of how the resources are physically arranged, the logical topology of the network and the interactions of its members are considered in

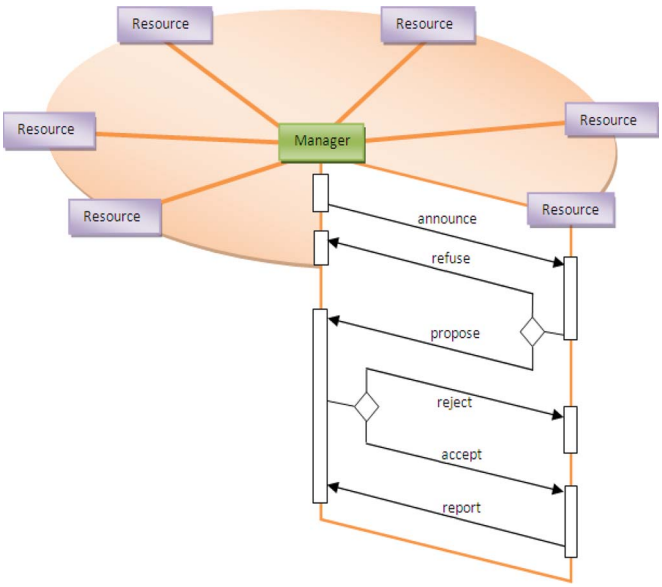


Fig. 2. Star model formed with CNP-based interactions, simplified from [45].

this paper. Alteration in the network architecture, interactions between entities, or even behaviors of each individual agent could influence the overall system performance. In this regard, other models for job allocation will be introduced in the following subsections. Initially, the details of rules in the star model will be defined.

In general, this task allocation model with CNP corresponds to the laws of typical market-based scheduling mechanisms [6], [21]. Nevertheless, in this paper, after receiving bids from the contractors, when the manager processes the bids and decides about awarding contracts (allocating tasks), the following rules are used.

1) IF more than one resource can complete a task before its due time, THEN the resource with the lowest cost will be chosen. In such a case, IF more than one resource has the same lowest cost, THEN the resource that can start earlier has priority.
2) IF only one resource can complete a task before its due time, THEN this it is chosen without any cost consideration.
3) IF no resource can complete a task before its due time, THEN the resource that can start earlier has priority, without considering the resource cost.

The CNP-based star model is a simple and well-known method of task allocation. However, it is still too centralized with only one manager, and does not allow diversity in agents [31], [47]. In the following subsection, an enhanced more flexible model is considered.

B. P2P Model

Applying the CNP in the star topology implies having one central manager. Although the centralized control as seen in the star model adds a global view to the system, it has a major drawback: If the unique manager of the system fails to operate properly, the whole task allocation process will break down [31]. Distributing the management and control of task

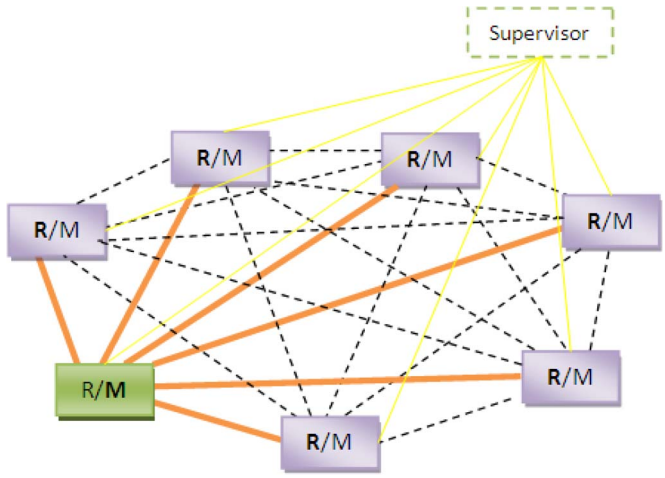


Fig. 3. P2P model, using CNP.

allocation in the system is therefore a proper strategy that results in a weblike P2P structure with multiple managers where each implements CNP. Devising a central supervisor, however, can keep the advantages of centralized systems, specifically its global view, as an important issue in small networks [31]. Manufacturing job shop applications (e.g., shop floor task allocation) are normally small networks of limited resources to be scheduled.

All resource agents in the star model are then connected to one another to produce a P2P model, as shown in Fig. 3. Here, no single central manager or broker exists, where each of the peers can generally be of the same status and can have the same capability and control right.

This means that any resource can be a manager too (as considered in this paper). In contrast with the star model, such P2P model is highly fault tolerant and robust due to redundancy of autonomous resource/manager (R/M) agents [3]. For the purpose of central coordination among the managers, a higher level supervisory agent is added here to the system.

As a first step, the supervisor groups the tasks in terms of their dependences and sends each group to an R/M agent. Depending on the number and dependence of tasks, each R/M agent could receive zero, one, or more than one task group. Then, each manager agent that possesses at least one task group uses CNP in conjunction with the relevant rules mentioned in the star model for negotiation and task allocation to all other R/M agents. The final schedule is then produced. Fig. 4 shows the algorithm used for this initial grouping and distribution of the incoming set of tasks among peers. In the algorithm, the condition of sequential manufacturing processes for production of a part is considered. Hence, the tasks are not necessarily independent, and some are prerequisite to others. This reflects the manufacturing process plan (e.g., the sequence of milling, boring, and turning operations in machining different features of a part).

The models presented up to this point are based on CNP which is a powerful standard protocol for interaction and task allocation. CNP is nevertheless not the only solution for this problem. By considering the notion of topology in a MAS, other models of agent interaction and job allocation may be proposed. A feasible classical network arrangement that could

```
set n to peers count
while task list is not exhausted do
  repeat for each integer from 1 to n
    if there is any task in the task list
      then read next task from task list
           add next task to orderList[n]
           remove next task from task list
           find all dependent-tasks for next task
           add dependent-tasks to orderList[n]
           remove all dependent-tasks form task list
    endif
  endrepeat
endwhile
```

Fig. 4. Algorithm of initial grouping and distribution of the tasks.

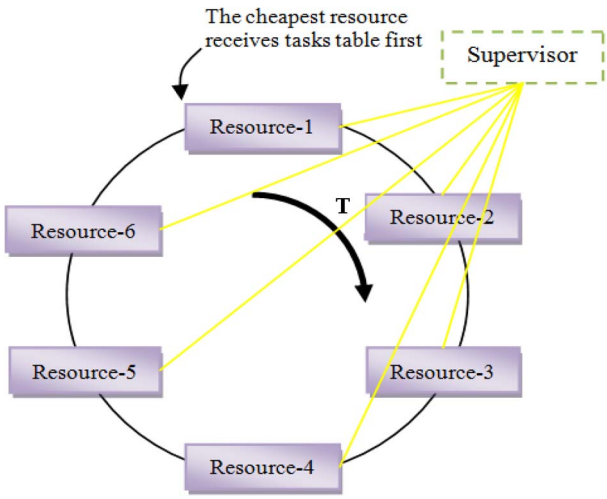


Fig. 5. Ring model.

be considered in the manufacturing task allocation problem is ringlike topology, which will be presented next.

C. Ring Model

Resource agents could be arranged to form a ring, as shown in Fig. 5. Here, there would be no manager agent as in the previous cases. A higher level supervisory agent is in charge of coordination similar to the P2P model. The main problem of classical ring topology is that the failure of one network member shuts down the entire network. However, here, the role of supervisor precludes such condition.

Upon arrival of a manufacturing order (set of tasks), a table of tasks to include all their specifications is created. The tasks are sorted in the table according to their priority, which is determined by predefined rules and user inputs. The supervisor agent successively circulates the task table among the resource agents and monitors it. The resource agent with the lowest operating cost is the first network member to receive the table. The agent that holds the table reviews all the remaining tasks in it, which are already sorted by the highest priority, and identifies the ones that match its technical capability. From the identified tasks, it then picks those that can perform within their due time, and adds them to its local schedule (selfish and greedy behavior). Then, it begins to execute the task with the highest priority.

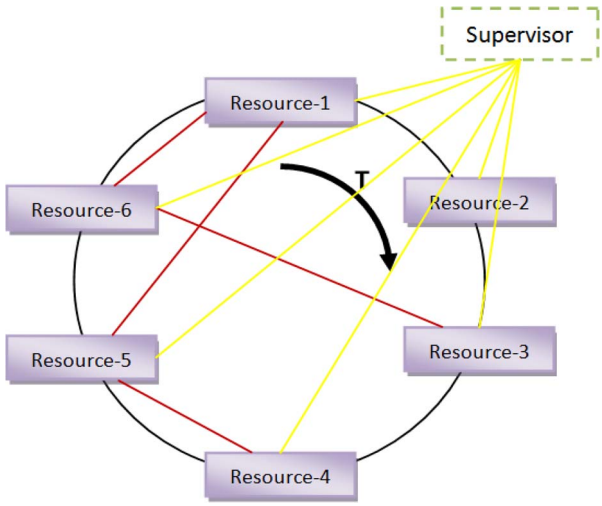


Fig. 6. Modified ring model.

The resource agent leaves a proposal for the tasks that match its technical capability, but is unable to meet their due-time requirement. On receiving the table, the next agent, if also unable to satisfy the due time, compares the left proposal with that of its own, and decides which one should be kept in the task table for further circulation (the worse proposal will be omitted). Each resource agent has its local schedule in which the IDs, together with all other attributes of the tasks undertaken, or the tasks it has offered a proposal for, are recorded. The table will be passed on to the next agents until all tasks are allocated.

The introduced ring model is a new approach to the task allocation problem using the ring topology which drastically differs from CNP and its variations, although it still uses the bidding mechanism to a limited extent. The lack of connection between two nonconsecutive agents in the ring, however, may simply reduce the capacity of the model. Accordingly, in the next subsection, a modified ring model is introduced.

D. Modified Ring Model

By improving the ring topology with advantages of P2P interactions, another topology model could be created. In this combination, referred to as modified ring in this paper, the structure and basic protocol are similar to that of the ring topology, but the agents can interact with one another through ACL messages in special situations, as shown in Fig. 6. For instance, when an agent has replaced the previous proposal by one of its own, it will notify the agent that had set the previous proposal to update its local schedule. The algorithm of the modified ring topology is shown in Fig. 7. In the algorithm, the condition of sequential manufacturing processes for production of a part is considered. Hence, the tasks are not necessarily independent, and some are prerequisite to others.

So far, four models for agent-based task allocation have been elaborated. The models based on CNP have already been widely used by researchers in academic and industrial applications, although different implementations vary in details. The models based on ringlike configuration for resource agents have been developed in this paper as alternative options in the task allocation problem. The models will now be evaluated and compared

```
supervisor extends agent
   sortResources (criteria)
   sortTaskTable (priorities)
   issueTaskTable (first resource)
resource extends agent
   while processTaskTable ( ) != 0 do
      if processTaskTable ( ) = 1 then takeTask (taskTable)
      endif
      if processTaskTable ( ) = 2 then proposeDueTime (taskTable)
         replacePreviousDueTime (taskTable) return outbade resource
         inform (outbade resource)
      endif
      sendTaskTable (next resource)
   endwhile
   if resource fails then return task
      supervisor:: sortResources (criteria)
      supervisor:: sortTaskTable (priorities)
      supervisor:: issueTaskTable (first resource)
   endif

Comments:
Criteria chosen are the resource operating cost in ascending order
and its activity status
Priority is a function of the inverse of task slack time
0 means that task table is empty or resource fails
1 means that resource can do task before the original due time or
resource has an existing proposal for the same task on the task table
2 means that resource cannot do task before the original due time and
resource has a proposed due time that is better than the previously
proposed due time for the task
```

Fig. 7. Algorithm of the modified ring model.

with one another in order to identify their specific advantages. The evaluation will require an appropriate tool for experimentation and determination of measures of performance. For this purpose, an agent-based simulation system will be presented in the next section, while performance indicators will be defined and measured through experimentation in Section V.

IV. Simulation System

Different methods can be considered to have a simulation tool for examination and comparative evaluation of the models explained earlier. In this section, the requirements and implementation of an appropriate system will be presented. The task allocation models are agent-based and have discrete time events. Therefore, the two concepts of discrete event simulation and agent-based simulation are relevant, where the latter is an extension of the former. Agent-based simulation is a kind of parallel discrete event simulation due to the use of agents as entities that can act concurrently [48]. It is also known as the most suited means of MAS validation [49].

There is no commercial multipurpose package for agent-based simulation to flexibly accommodate implementation of models and protocols described in this paper, although commercial software for discrete event simulation of manufacturing processes/environments is available. Java is known as a leading object-oriented programming language for agent development due to its useful relevant features. Furthermore, a number of major platforms/toolkits such as NetLogo, MASON, Swarm, and Repast are available. Among them, Swarm and Repast are directly developed for simulation purposes [1].

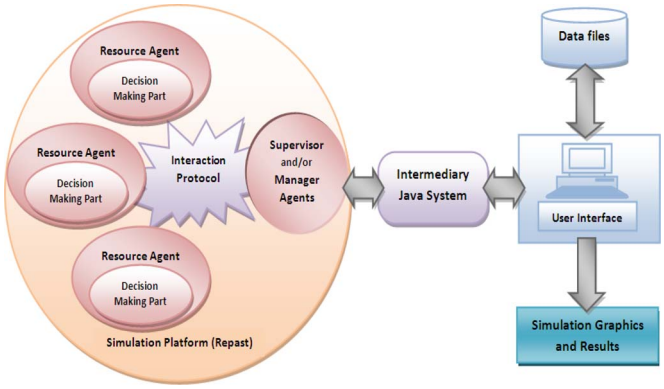


Fig. 8. Overall system architecture.

Repast, as open-source software, is the most complete Java-based simulation platform with good execution speed, although it has deficiencies in some aspects such as documentation [50]. In this paper, Repast is used to build the simulation system. It provides integrated library of classes that assists modelers in an agent-based simulation, and is robust, extensible, and easy to use [51]. Repast uses a unit of time named "tick" for simulating discrete events, which will be required for the models of this paper.

A. System Requirements and Architecture

The simulation system aimed for this paper is required to take the set of tasks and available resources, visualize and monitor them in the four agent-based models of task allocation (described in Section III), complete the scheduling of the tasks, calculate several performance measures (time, cost, resource utilization, etc., as will be discussed in Section V), and report the obtained task allocation schedule and performance parameters of each model. The user will then be able to compare the four topology models after each simulation run.

The overall system architecture is shown in Fig. 8. In this architecture, there is a simulation platform consisting of agents in Repast environment. Other pieces of Java codes acting besides the Repast environment are shown in Fig. 8. Resource agents have two components: one for decision making and scheduling issues, and the other for operation (task execution) in the simulation environment. The two parts act in parallel, allowing the resource agent to execute the tasks while it is negotiating or making a decision. This makes possible to implement the modified version of CNP as discussed in Section III. The second type of agents includes those acting as supervisor or manager, as applicable. Such agents have rule-based inference and control function only, without an operational part. They regulate the interaction and collaboration of the autonomous operational agents in the system and are "explicit control entities" [52], which have global knowledge of tasks and available resources.

B. System Implementation and Functionality

The data relating to resources and tasks to be carried out are organized in data files, which are read through user interface. The user selects which type of data is to be used for a specific simulation, and has other options in the interface data input,

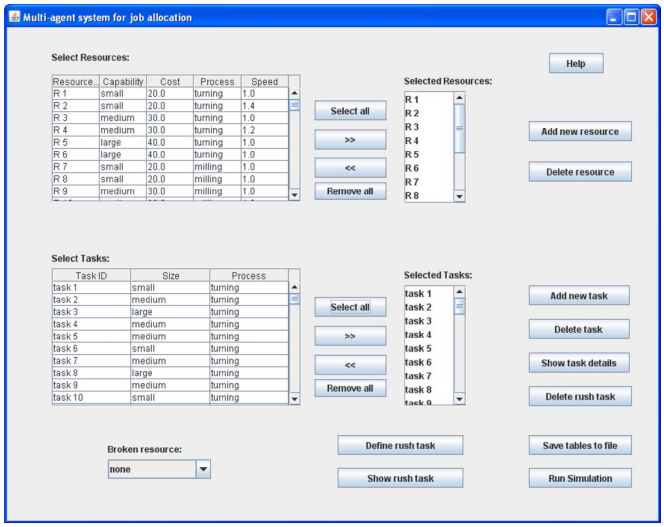


Fig. 9. User interface for data input, showing a case of shop floor task allocation.

as partly shown in Fig. 9. Each task (namely, part to be produced) and resource (namely, a manufacturing machine) has their own specifications, as what generally exist in a typical manufacturing.

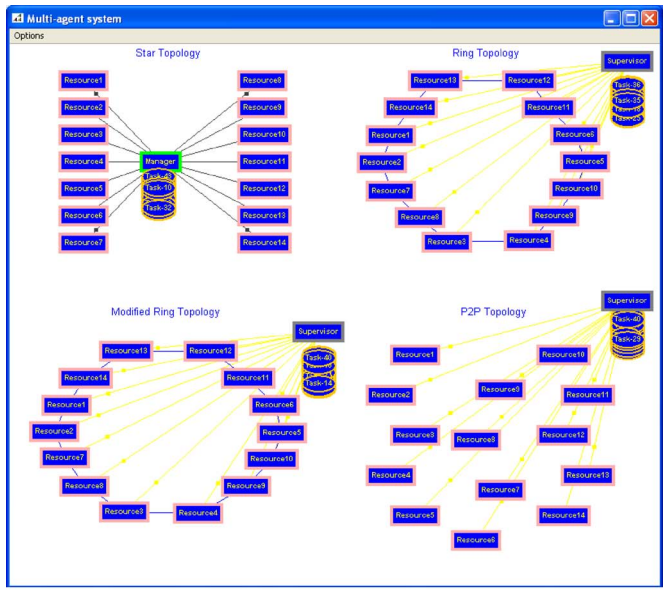
Here, the task data include its identification, part size, the required manufacturing process (milling, turning, drilling, or assembly operations for instance), standard operation time for a routine production or estimated time for a new one, task due time and its earliest possible start time (depending on availability of raw material, etc.), task penalty costs if the due time is passed, and any dependence of the tasks according to the relevant process plan (as one may be a prerequisite of another). Machine data, however, include its identification, the manufacturing processes it can perform, geometrical constraints, operational speed, cost of operation, and idleness. Each resource agent is a specialist on certain type of tasks and bid for a task only if it matches its capabilities. The matching is done by a rule-based mechanism.

By using graphical facilities embedded in Repast, visualization and monitoring of the agent-based simulation process are made possible for the user. Snapshots of a sample run are shown in Fig. 10(a) and (b), depicting the status of the task allocation process. These figures also give a better perception of four models and different interaction types in each.

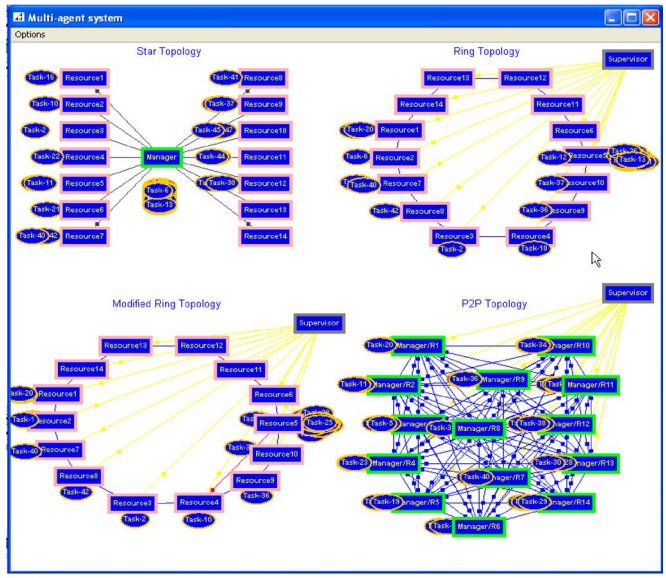
Another output of user interface is graphical presentation of final results after simulation ends. The user has the opportunity to see the complete schedule of each resource in the form of Gantt charts which show allocation and sequence of execution of tasks. A sample of such output is shown in Fig. 11. The task color changes by exceeding the respective due time, and penalty cost starts to be calculated. Overall, results including time, costs, and resource utilization are shown as output. The results and their implications in a real world will be discussed in the next section.

V. Evaluation

The system described in the previous section, together with the data from an industrial case study, will be used to conduct



(a)



(b)

Fig. 10. (a) Snapshot of the simulation graphics just before the allocation process starts. (b) Snapshot of the simulation graphics during the allocation process.

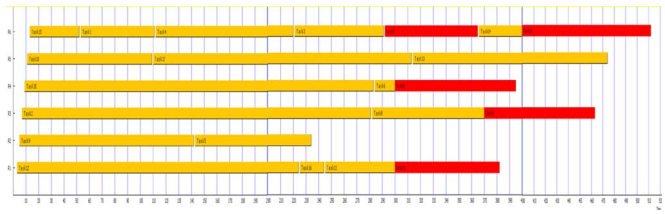


Fig. 11. Sample Gantt chart of resources at the end of simulation.

task allocation experiments and compare the performance of the models described in Section III. Before performing the experiments, however, evaluation of the experimentation platform is presented, and a set of performance measures for the models are defined.

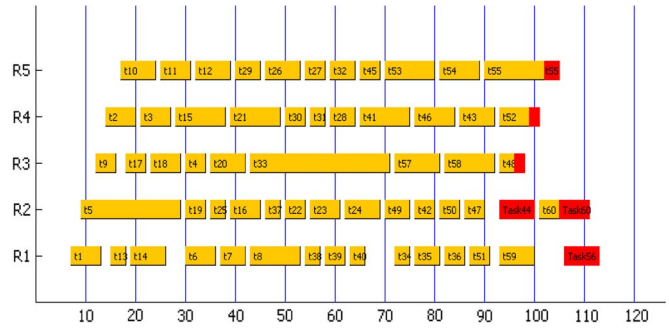


Fig. 12. Output of the system for an evaluation test case.

A. Evaluation of the System

In addition to data validation in the system, different functionalities have been checked and verified during the system's development and at its completion using simple data sets. Furthermore, a validation test has been conducted using the manufacturing task allocation data provided by Vancza and Markus who used a bidding mechanism among five machine tools in a study of shop floor job allocation [53]. In this paper, there are 15 parts to be machined, each having four processes of boring, drilling, or milling types. This means that a total of 60 tasks exist in this case study, in which every four tasks have dependence according to the process plan of the respective part. Each part arrives at a given time, which means that the earliest start time is not zero for them. Their due time is also given, and a penalty is imposed in case the due time is not satisfied. The data were compiled as the input to run the software for the CNP-based models that is comparable to the bidding mechanism in this case study. The results, as depicted in the Gantt chart of Fig. 12, showed a close correspondence of the P2P model with the aforementioned work in terms of distribution of tasks with the resources and the total time to complete them.

B. Defining Performance Indicators

A number of performance indicators are defined here. Many criteria can be used for evaluation of the task allocation models in manufacturing domain [11]. Some of them are time relevant (e.g., total time of completing tasks and number of tasks which finish with delay, rate of production, etc.), and some are to assess optimum management of the resources (e.g., machine utilization and balance between them). Cost is an important parameter which is not independent of time, but has certain features with no direct correlation to time. It will thus be considered as a separate evaluation criterion.

Defining appropriate performance indicators depends on the type of the manufacturing scenario and the data available from it. For example, the rate of production or the number of products shipped in a certain period is not as critical as the lead time for make-to-order products. Considering all the points discussed, the following quantitative parameters are calculated from output data of the simulations as performance indicators in this paper.

1) *Total time*: This is the time elapsed to complete a manufacturing order (set of tasks), and contains any time spent on scheduling as well as operations until the last resource finishes the last task. It is also referred to as lead time.
2) *Costs*: Contains three major cost elements of the resources (machines) in total. The first element is the cost when a machine is *busy* with a task. This is calculated by the rate of busyness which depends on depreciation and running costs of each machine and the duration of operations including setup times. The second element is *penalty* cost if a task passes its due time. Rate of penalty for each task is defined in the manufacturing order. Penalty cost has also an indication of tardiness. Finally, the third cost element is relevant to *idleness* (i.e., nonoperating) status of machines.
3) *Utilization*: Defined as the percentage of processing time during the total time of executing an order. It will be shown by busy/idle percentage of the machines.

C. Experiment With a General Case of Manufacturing Shop Floor

Industrial case study data involving a large gas turbine production shop floor have been used to support evaluation of the models in this paper. Incoming materials to the manufacturing shop floor are cast, forged, or welded parts that are to be machined using computer numerical control milling and turning machine tools, preassembled in some stages, and finally dispatched to assemble the final product. Each product has a set of parts to be produced in the shop floor. The manufacturing order is received at the workshop to make one or more turbines either in a make-to-order or make-to-stock fashion. Table I gives the data in the case study. The study includes six turning, six milling, and two assembly stations, each with different sizes, costs, and capabilities that cover typical demands.

Fig. 13 shows the results of each model in terms of performance indicators defined earlier. Fig. 13(a) shows the total time elapsed, in which the modified ring offers the minimum time, while the ring model offers the highest time taken to complete the order. Fig. 13(b) shows the three components of the total cost as percentages. Here, the total minimum cost belongs to the P2P, and the modified ring is in second place, due to higher penalties paid. Similarly, the utilization percentages in Fig. 13(c) show that the modified ring and P2P are better than the other models with average busy time higher than their idleness (more than 50% busy). The modified ring exhibits the best utilization performance (52.8% busy compared to 51% of the P2P). The standard deviations (SDs) of the busy percentages within each model were also calculated, and showed no significant difference among the models (all around 20%). This means that the pattern of busy time distribution over the machines is similar. In summary, the models P2P and modified ring are both quite promising in this general case of machining shop floor.

D. Further Experiments for Special Cases

Special cases occur in practice, in which one of the models might be considerably more suitable. In this section, a number of them are observed, based on the data taken from the

TABLE I
TASKS DATA FOR THE GENERAL CASE OF MANUFACTURING SHOP FLOOR

task ID	start time	operation hours	due time	size	penalty charge per hr	manufacturing process	pre-task
1	0	58	200	small	5	turning	none
2	0	274	290	medium	10	turning	none
3	0	70	300	large	5	turning	35
4	0	108	260	medium	10	turning	36
5	0	8	370	medium	5	turning	26
6	0	65	250	small	5	turning	none
7	0	93	300	medium	10	turning	37
8	0	73	280	large	10	turning	38
9	0	181	370	medium	10	turning	39
10	0	98	190	small	10	turning	40
11	0	98	210	small	10	turning	41
12	0	137	300	small	5	turning	42
13	0	9	440	medium	5	turning	27
14	0	9	300	small	5	turning	28
15	0	10	300	small	5	turning	29
16	30	221	300	small	10	turning	none
17	0	160	520	large	10	turning	43
18	0	146	400	large	10	turning	44
19	0	39	180	medium	10	turning	45
20	20	19	300	small	5	turning	none
21	0	203	330	large	10	turning	46
22	0	229	330	medium	10	turning	47
23	0	82	400	medium	5	turning	30
24	0	39	500	large	10	turning	31
25	0	71	530	medium	5	turning	32
26	0	100	360	small	5	assembly	4
27	0	55	430	small	5	assembly	9
28	0	55	250	small	5	assembly	10
29	0	55	270	small	5	assembly	11
30	0	120	300	small	5	assembly	19
31	0	120	450	large	5	assembly	21
32	0	120	450	small	5	assembly	22
33	0	76	280	small	5	milling	1
34	0	132	420	medium	10	milling	2
35	0	73	230	large	5	milling	none
36	0	146	150	medium	10	milling	none
37	0	151	200	medium	10	milling	none
38	0	180	200	large	10	milling	none
39	0	183	190	medium	10	milling	none
40	0	83	90	small	10	milling	none
41	0	99	110	small	10	milling	none
42	0	130	150	small	5	milling	none
43	0	320	350	large	10	milling	none
44	0	238	250	large	10	milling	none
45	0	133	140	medium	10	milling	none
46	0	110	120	large	10	milling	none
47	0	95	100	medium	10	milling	none

industrial case study. First, suppose that no penalty is charged for passing the due times. Under such conditions, the penalty portion in the cost bar chart of Fig. 13 would be removed. This

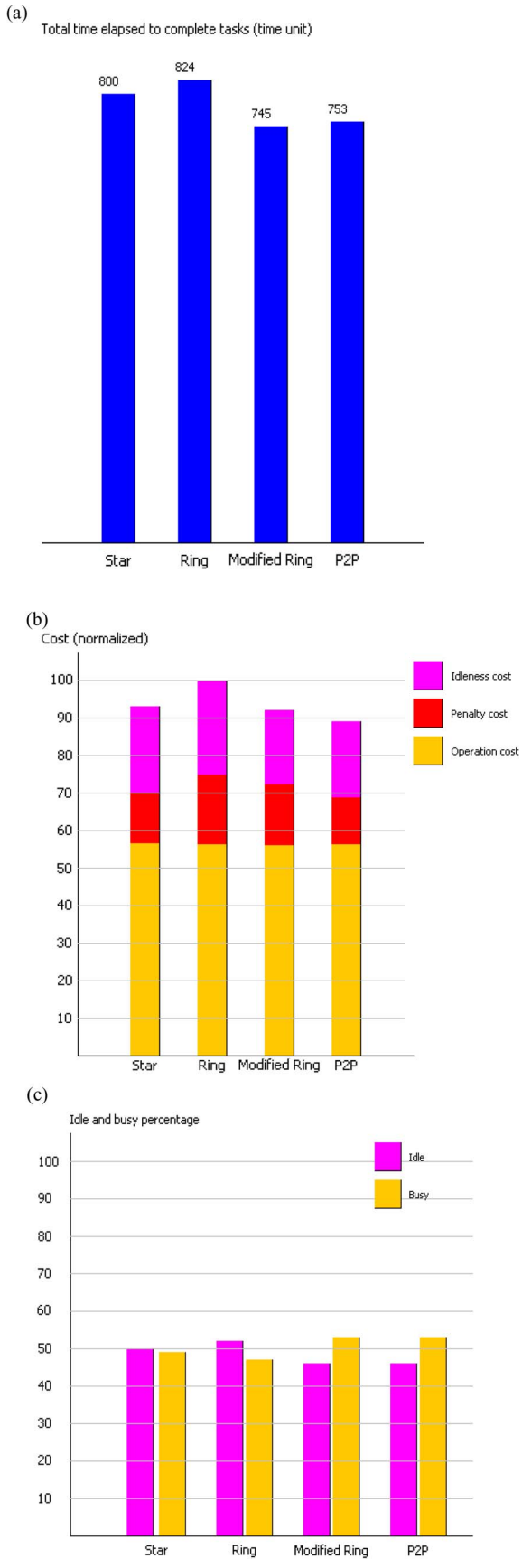


Fig. 13. Results of a general case of manufacturing workshop.

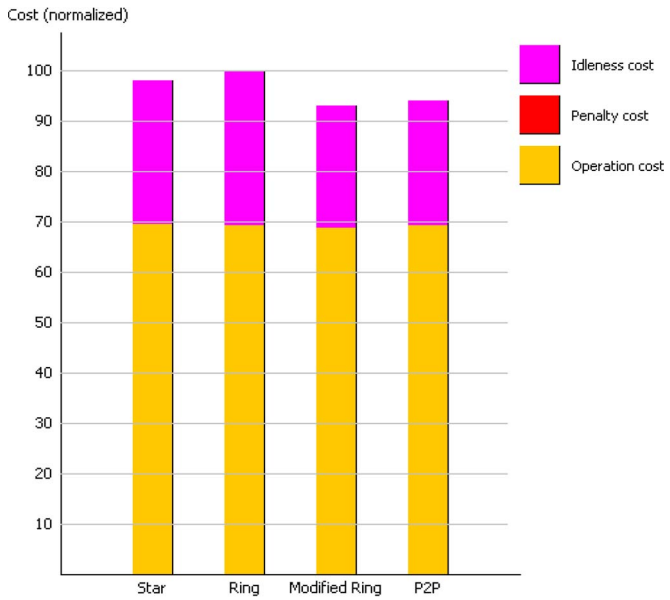


Fig. 14. Comparison of costs when no penalty is charged.

results in the modified ring to exhibit the lowest cost compared with the four models, as shown in Fig. 14.

In the next attempt, a relaxation is made on the due times, extending them by 100 units of time which is around 20% of the maximum due time in the general case of the previous subsection C. This compares the models with regard to time strictness. Results are shown in Fig. 15. Here, the P2P model shows the best performance, while the star model is second with a small difference. This could be due to the selfish and greedy behavior of the ring agents, as described in Section III. The behavior makes the lower cost agents to pick any task from the task table as long as they can carry it out within the due time. However, this is done without considering a uniform distribution among the resources. As a consequence, more idleness of the higher cost resource agents and a busy schedule for the lower cost ones will occur. This may finally lead to worse overall results in terms of time and cost for the system.

In order to observe the results when there are jobs without dependence to one another, a part of machining tasks (18 turning processes) from the first case of the previous subsection C was selected. Fig. 16 shows the results in which the modified ring outperforms in all indicators and the star and P2P performances are very close to each other. This is a rather special case, but can occur particularly when there are separately managed clusters of machinery of one process type.

E. Rush Task Experiment

In order to compare the models when an unexpected event occurs, rush tasks have also been introduced to the system in random various times during execution of the manufacturing order in the case of independent jobs. This included tasks requiring urgent attention with priority to all other tasks, except the ones that already started their operations. The tasks under operation are not therefore interrupted, but the others in the local schedule of resource agents are recalled and a

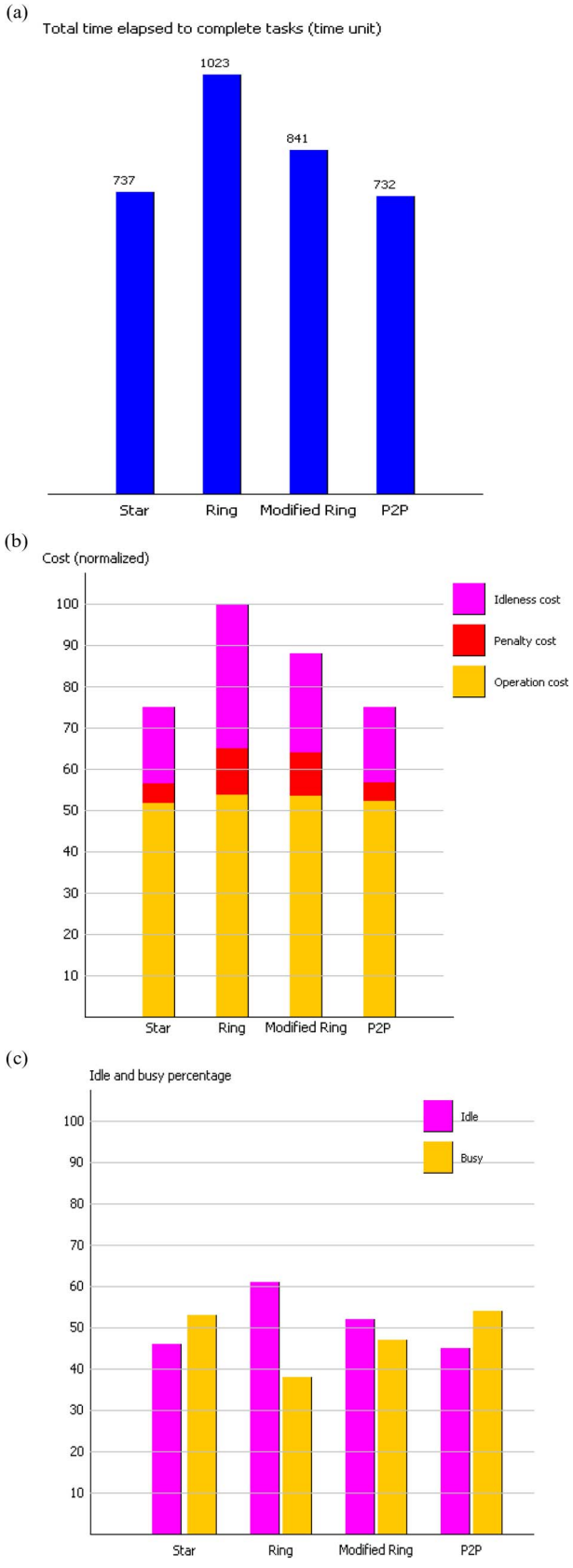


Fig. 15. Results when due time is extended.

fresh allocation process is run. The results, shown in Table II, suggest that the modified ring has the lowest lead time on *average* (Ave), followed by the P2P. However, the CNP-based

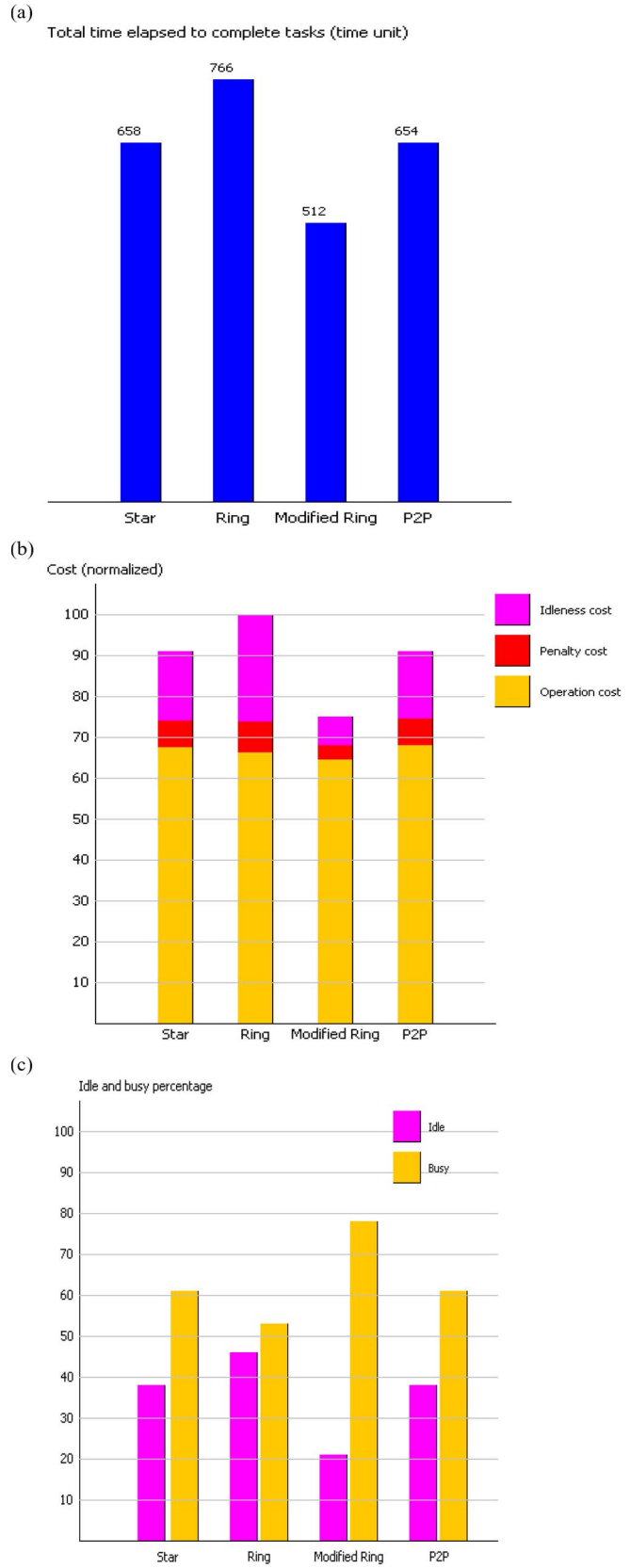


Fig. 16. Results for independent tasks (18 tasks, only turning, using 6 turning machines).

models (star and P2P) have better SD. This suggests that less variation in outputs in different runs is exhibited. The lower variation implies that models based on CNP are less sensitive to occurrence of unexpected events requiring priority in operation.

TABLE II
LEAD TIMES IN DIFFERENT RUSH TASK EXPERIMENTS

	Star	P2P	Ring	M. Ring
	624.0	625.0	686.0	633.0
	629.0	625.0	613.0	586.0
	629.0	625.0	606.0	589.0
	629.0	625.0	606.0	590.0
	624.0	625.0	606.0	633.0
	629.0	625.0	606.0	589.0
	629.0	625.0	589.0	589.0
	629.0	587.0	720.0	606.0
	621.0	580.0	778.0	606.0
	621.0	580.0	590.0	606.0
	584.0	580.0	589.0	512.0
	621.0	580.0	586.0	613.0
	584.0	580.0	737.0	621.0
Ave	**619.5**	**604.8**	**639.4**	**597.9**
SD	**16.1**	**22.8**	**66.4**	**30.6**

F. Evaluation of the Performance of Agents

By inductive reasoning, some of the empirical results in subsections C–E are explained. The combined effects of the topology and protocol on the decision-making scheme of the agents may be indicated by the communication efficiency, the degree of decentralization, the probability of overall system failure, task allocation efficiency, and due-time deviation, as shown in Table III.

1) Communication efficiency is denoted here by a ratio, determined by the number of messages exchanged, resulting into selected agents, divided by the total number of messages exchanged among all agents that were contacted.
2) The degree of decentralization is denoted by the percentage of work allocation duties that were carried out by the resource agents, rather than by the supervisor or the manager.
3) The probability of overall system failure is a function of the probability of agent and/or manager failures.
4) Task allocation process efficiency is given by a ratio, determined by the number of tasks allocated in one communication round with the central company, divided by the total number of tasks.
5) Due-time deviation is given by the difference between the due time proposed by the resource agents and the optimal feasible due time that the resource agents could have proposed.

In Table III, k is the number of resource agents that reached the fourth step in the CNP of Fig. 1, n is the number of resource agents that stopped at step 3, m is the number of agents that stopped at step 2, p is the number of agents that did not respond, T is the total number of agents contacted, x is the number of agents that proposed new due times, α is the number of tasks allocated, and β is the total number of tasks ordered.

TABLE III
PERFORMANCE OF THE MODELS FROM AN AGENT PERSPECTIVE

Performance Indicators	Star Topology	P2P Topology	Ring Topology	Modified Ring Topology
Communication efficiency (T = k + n + m + p)	$\frac{4k}{4k+3n+2m+p}$	$\left(\frac{4k}{4k+3n+2m+p}\right)^{\tau}$	$\frac{k+2}{T+2}$	$\frac{k+2}{T+3+(x-1)}$
De-centralisation percentage	50%	100%	57%	63%
Probability of overall system failure	≥ probability of failure of manager	≪ probability of failure of an agent	= probability of failure of an agent	< probability of failure of an agent
Task allocation process efficiency	$=\frac{1}{\beta}, \beta > 0$	$=\frac{1}{\beta}, \beta > 0$	$=\frac{\alpha}{\beta}$	$=\frac{\alpha}{\beta}, \beta > 0$
Deviation of proposed due time	Low	Low	Very high	Medium

TABLE IV
EXAMPLE OF THE EVALUATION OF THE MODELS

Performance Indicators	Star Topology	P2P Topology	Ring Topology	Modified Ring Topology
Communication efficiency	100%	100%	100%	100%
Task allocation process efficiency	5%	5%	100%	100%

For example, if all of the, for example, ten available agents are selected, and they all can produce the tasks before the due time, while the modified ring topology allocated all of, for example, 20 tasks in one circulation of the task table, then $k = T = 10$, $n = m = p = 0$, $x = 0$, and $\alpha = \beta = 20$. These results are shown in Table IV.

In this example, for the CNP-based star model and the P2P model, the 40 messages exchanged with the 10 agents resulted into all the agents being selected. In the case of the ring and modified ring models, the exchange of 12 messages resulted into all agents being selected. All four models achieved a 100% communication efficiency based on their respective algorithms. This example also presents the task allocation process efficiency. In the case of the star and P2P models, one task is allocated for each communication round with the central company (which consists of four messages per agent, as shown in Figs. 1 and 2). Thus, the task allocation efficiency is 1/20. In the case of the ring and modified ring models, 20 tasks are allocated for each communication round with the central company (which consists of two messages, as shown in Fig. 7). The task allocation efficiency is 20/20.

Tables III and IV give a different perspective to the empirical results. For instance, in part C of this section, the modified ring gives the minimum time due to the excellent efficiencies of the communication and task allocation process, and a medium deviation of the proposed due time from the best due time it could have proposed, as shown in Fig. 13(b). This deviation is reflected by the penalty cost being higher than that of P2P. The deviation occurred due to an agent temporarily holding capacity for a task which was eventually awarded to a different higher bidding agent, causing a snowball effect on the proposed due time, of the former agent, for the next tasks. Allowing the resource agents to revise their proposed due times when outbid reduces the due-time deviation. Furthermore, in part E of this section, it is implied that the P2P and the star model are least sensitive to occurrence of unexpected events. This is explained by the relatively poor communication efficiency and task allocation process efficiency of both models.

In summary, none of the compared models outperforms with all kinds of input data. Nevertheless, it is evident from the results that the simple ring model is the weakest performer and has no advantage in the majority of cases. An added ability of P2P information exchange and cooperation to the ring, however, has created a powerful model of modified ring. It is one of the two best performers. The other outstanding model is the P2P which is CNP-based. The higher performance of the P2P model is attributed to its higher level of distribution and redundancy. Although the CNP-based star model generally performs well, it shows no major advantage over the P2P. The P2P interaction capability in the modified ring and P2P allows them to fully exploit decentralization which is a key feature in the agent-based systems. The application of the modified ring model will have potential contribution in agent-based environments, such as the formation and operation of manufacturing networks, in which a number of small and medium enterprises (SMEs) work together to fulfill a large order [54], [55]. The resources in such a network are SMEs with characteristics comparable to machines in a workshop. This paper is a comparative study in the field of agent-based scheduling. The relative performance value and position of the proposed models in this paper are determined in relation to those of the established CNP-based star and P2P models, which are marketlike methods. Comparison with the conventional scheduling practice in the real world is also the subject of the future part of this paper.

VI. CONCLUSION

In this paper, four agent-based models for task allocation in manufacturing shop floor have been presented and compared by using Java-based simulation software developed as the test platform. The two models referred to as star and P2P, respectively, used the established and popular CNP, while the other

two models introduced in this paper have been built using novel architecture and algorithms. Initially, the prominent position of the agent-based scheduling within the broad area of scheduling has been discussed.

Experiments were conducted using real manufacturing data to test the performance of these models. Lead time, cost, and resource utilization have also been used as the performance criteria. The results show that, in most cases, the proposed modified ring and CNP-based P2P models give superior performance compared with the star and ring models. The new modified ring model, with its protocol developed in this paper, is therefore a serious competitor to the CNP-based models.

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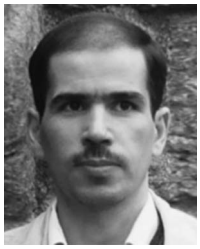
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